

Weekly Spotify Wrapped

M122E Test Protocol

1. Test Environment

Item	Value
Operating system	Windows
Python	Python 3.14.3
Test framework	pytest
Project folder	spotify_weekly_wrapped
Main command	python main.py
Test command	python -m pytest

2. Automated Unit Tests

ID	Test Case	File	Expected Result	Status
T-01	Load valid config with email disabled.	test_config.py	Config object is created.	Passed
T-02	Detect missing Last.fm values.	test_config.py	ConfigError lists missing values.	Passed
T-03	Require Gmail values when SEND_EMAIL=true.	test_config.py	ConfigError mentions Gmail fields.	Passed
T-04	Support UTF-8 BOM .env files.	test_config.py	Config loads correctly.	Passed
T-05	Prevent duplicate database rows.	test_database.py	Duplicate play is skipped.	Passed
T-06	Fetch tracks only inside requested period.	test_database.py	Only matching tracks are returned.	Passed
T-07	Update duplicate rows with Spotify metadata.	test_database.py	Metadata is saved.	Passed
T-08	Calculate statistics and listening time.	test_report_generator.py	Stats match sample data.	Passed
T-09	Generate HTML report.	test_report_generator.py	Report file is created.	Passed
T-10	Compare current and previous week.	test_report_generator.py	Percentage changes are calculated.	Passed
T-11	Normalize Last.fm tracks and handle API errors.	test_lastfm_client.py	Clean data or clear error.	Passed
T-12	Enrich tracks with Spotify data.	test_spotify_client.py	Spotify metadata is added.	Passed
T-13	Handle Spotify playlist/API errors.	test_spotify_client.py	Errors are clear and controlled.	Passed
T-14	Send multipart email and validate missing values.	test_email_sender.py	SMTP is mocked successfully.	Passed
T-15	Create OAuth URL and update .env values.	test_spotify_setup.py	Setup helper works safely.	Passed
T-16	Test main workflow and optional playlist failures.	test_main.py	Correct exit codes and fallback.	Passed
T-17	Verify project structure and tests for every module.	test_project_structure.py	Required files exist.	Passed

3. Latest Automated Result

37 passed

4. Manual / Live Tests

ID	Test Case	Expected Result	Actual Result	Status
L-01	Missing .env handling	Clear missing-value error.	Missing values were listed clearly.	Passed
L-02	Last.fm API fetch	Tracks are loaded.	Real listening data loaded successfully.	Passed
L-03	Spotify enrichment	Metadata is added where possible.	Spotify enrichment completed.	Passed
L-04	SQLite duplicate handling	Second run skips duplicates.	Duplicates were skipped.	Passed
L-05	HTML report generation	Report is saved in output/.	Dated report was created.	Passed
L-06	Gmail email delivery	Report email is delivered.	Email arrived successfully.	Passed
L-07	Playlist error handling	Playlist errors do not stop report.	HTTP 403 was handled.	Passed
L-08	Mobile email preview	Email is readable on phone.	Dark layout with covers worked.	Passed

5. Edge Cases Covered

- Missing environment variables.
- UTF-8 BOM in .env file.
- Duplicate database records.
- Missing Spotify duration fallback.
- Missing previous-week comparison data.
- Spotify playlist permission error.
- Email sending failure path.
- Main workflow without real external services in unit tests.

6. Test Evidence

Loaded tracks from Last.fm.

Spotify enrichment finished.

Database saved new tracks and skipped duplicates on repeated runs.

Playlist update failed; continuing without playlist changes: HTTP 403 Forbidden

Report saved to output/weekly_spotify_wrapped_YYYY-MM-DD.html

Weekly report email sent to RECIPIENT_EMAIL.

7. Test Conclusion

The tests cover all application modules and the most important live workflow paths. The optional Spotify playlist error was handled correctly, which demonstrates that the script continues running even when a non-critical external service fails.